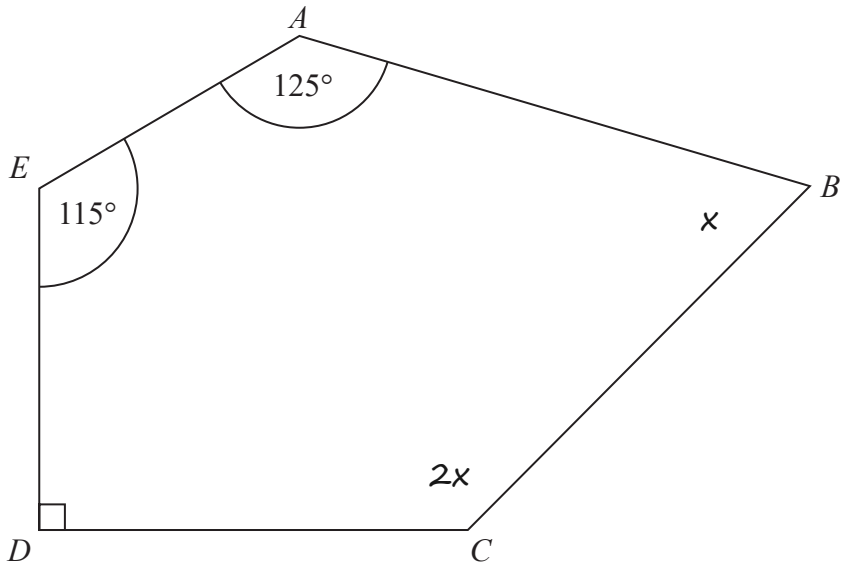
8 *ABCDE* is a pentagon.



Angle *BCD* = 2 × angle *ABC*

Work out the size of angle *BCD*.
You must show all your working.

Interior angle sum = (5 – 2) × 180°

1 mark
Use the polygon angle-sum rule

= 3 × 180° = 540°

1 mark
Correct total for a pentagon

Angles B and C total 540° – 125° – 115° – 90° = 210°.

1 mark
Subtract the three known interior angles

B : C = 1 : 2. The table gives their angles in degrees.

B		C		Sum
1	×70	2	×70	3 ×70
70		140		210

1 mark
Share 210° in the ratio 1 : 2

Angle BCD = 2 × 70° = 140°

1 mark
Answer for the required angle

140 °

(Total for Question 8 is 5 marks)